



L₁ Regularization for Robust Rainfall Prediction: A Case Study of Singapore's Meteorological Station Network Using Machine Learning

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Abstract

This study presents the development of a daily rainfall prediction model utilizing historical meteorological data from forty-four observation stations in Singapore, a region characterized by a tropical climate with high rainfall variability. The increasing frequency of short-duration, high-intensity rainfall events poses significant challenges for urban flood mitigation and hydrometeorological risk management, underscoring the need for accurate and robust forecasting tools. To address this, we conducted a benchmarking study comparing the predictive performance of three machine learning algorithms—Lasso Regression, XGBoost Regression, and Gradient Boosting Regression. The models were trained on numerical variables, including sub-hourly rainfall, daily rainfall, temperature, and wind speed. A comprehensive pre-processing pipeline was applied to the dataset, involving the cleaning of invalid values, imputation using monthly medians, outlier detection, date standardization, and the incorporation of temporal features. The models were trained on data from 1980 to 2023 and evaluated on a holdout set from 2024 to 2025 to ensure a realistic assessment of their predictive capabilities. Results demonstrate that Lasso Regression consistently outperformed the other models across all stations, exhibiting lower prediction errors, higher explained variance, and enhanced model stability. The superior performance is attributed to the L1 regularization mechanism in Lasso, which effectively mitigated multicollinearity and produced a more parsimonious model without compromising accuracy. These findings indicate that regularized linear regression is a highly effective approach for daily rainfall prediction from numerical meteorological data, holding considerable potential for enhancing early warning systems, flood mitigation strategies, and extreme weather risk management in tropical urban areas.



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1. Introduction

In recent decades, the increased frequency and intensity of extreme weather events have become a global concern, as climate change amplifies hydrological and atmospheric variability [1, 2]. Short-duration, heavy rainfall events are becoming more frequent in tropical regions, posing significant risks to infrastructure, transportation, and public safety. This trend necessitates more accurate weather forecasting, particularly in densely populated urban environments sensitive to microclimatic changes.

Singapore exemplifies this challenge. Its highly urbanized landscape induces significant microclimatic variability that complicates convective rainfall formation [3]. Factors such as the urban heat island effect, altered albedo, and building geometry can influence local convection patterns and extreme rainfall intensity [4]. Consequently, accurate rainfall

prediction is crucial for flood risk mitigation and adaptive urban planning.

While the Meteorological Service Singapore (MSS) operates a comprehensive network of weather stations [5], using the entire network for machine learning is not always optimal. Research indicates that a carefully selected, climatologically representative subset of stations can achieve comparable predictive performance while reducing model complexity [6, 7]. Accordingly, this study utilizes data from several key stations that best capture Singapore's rainfall variability.

Traditional rainfall prediction models, often based on linear statistics or numerical weather models, struggle with the high non-linearity of convective processes in the tropics, especially for high-intensity events [8]. Machine learning, particularly ensemble and boosting algorithms, offers a powerful alternative by effectively capturing non-linear relationships and complex interactions between

meteorological variables [9, 10]. These models excel at handling heterogeneous data, are robust to outliers, and offer flexibility in modeling complex dependencies. However, their performance is context-dependent, varying with local characteristics and data resolution.

Based on this background, this study aims to benchmark the performance of several machine learning algorithms for rainfall prediction using data from selected weather stations in Singapore. The primary focus is to evaluate each model's accuracy, stability, and generalization capability, specifically for heavy rainfall prediction. This approach seeks to identify the most effective model for a tropical urban environment, providing a scientific basis for developing more adaptive, data-driven operational forecasting systems.

2. Materials and Methods

2.1. Data Understanding

Daily meteorological data spanning from 1980 to 2025 were obtained from four observation stations in Singapore. The dataset comprises the following numerical independent variables: Highest 30-minute Rainfall (mm), Highest 60-minute Rainfall (mm), Highest 120-minute Rainfall (mm), Daily Rainfall Total (mm), Mean Temperature (°C), Maximum Temperature (°C), Minimum Temperature (°C), Mean Wind Speed (km/h), and Maximum Wind Speed (km/h). The Daily Rainfall Total was designated as the target variable for prediction.

2.2. Data Preprocessing

Data processing commenced with the consolidation of CSV files from all stations into a centralized repository. An automated pipeline was implemented to perform comprehensive data cleaning and standardization. The initial stage involved removing non-numeric characters from numerical columns to ensure data consistency. Missing values were then addressed through imputation using monthly medians, a method selected to preserve seasonal patterns in the data. Subsequent steps included standardizing date formats to YYYY-MM-DD, converting data types to float, and detecting outliers using the Interquartile Range (IQR) method to minimize the influence of erroneous extreme values. Finally, temporal features including Year and Month were engineered before the finalized dataset was saved.

2.3. Model Development

The dataset was partitioned chronologically into a training set (1980-2023) and a test set (2024-2025) to prevent temporal data leakage and ensure realistic evaluation of predictive performance. At each of the four observation stations, three distinct models were independently trained and evaluated.

Lasso Regression was implemented as a baseline model to establish linear performance benchmarks while performing feature selection through L1 regularization, thereby retaining only the most relevant predictors. To capture potential non-linear relationships and complex interactions among atmospheric parameters, two advanced ensemble methods—XGBoost Regression and Gradient Boosting Regression—were employed. These boosting algorithms

were selected for their capacity to handle data heterogeneity, model non-linear patterns, and provide greater predictive flexibility compared to regularized linear approaches

2.4. Model Evaluation

Model performance was evaluated using three distinct metrics: Mean Absolute Percentage Error (MAPE), Coefficient of Determination (R^2), and Mean Squared Error (MSE). These metrics were calculated for each model at each of the four stations to assess accuracy, error characteristics, and explanatory power from both scientific and operational perspectives. For a given station j (where $j = 1, \dots, J$) with N_j observations, the actual and predicted values are denoted as $y_i^{(j)}$ and $\hat{y}_i^{(j)}$, respectively. The mean of the actual values at station j is defined as:

$$\bar{y}_i^{(j)} = \frac{1}{N_j} \sum_{i=1}^{N_j} y_i^{(j)} \quad (1)$$

The evaluation metrics are computed for each model at each of the four stations to assess model accuracy and error characteristics at the station level, while providing scientific and operational justification for the selected metrics.

The Mean Absolute Percentage Error (MAPE) for station j is defined as:

$$MAPE^{(j)} = \frac{100\%}{N_j} \sum_{i=1}^{N_j} \left| \frac{y_i^{(j)} - \hat{y}_i^{(j)}}{y_i^{(j)}} \right| \quad (2)$$

MAPE presents the average absolute error as a percentage of the actual value, making it easy to interpret and useful for comparing performance across stations of different scales. However, MAPE has limitations in that it is undefined when $y_i^{(j)} = 0$ and gives excessive weight to cases with small $y_i^{(j)}$.

The Coefficient of Determination (R^2) for station j is given by:

$$R^{2(j)} = 1 - \frac{\sum_{i=1}^{N_j} (y_i^{(j)} - \hat{y}_i^{(j)})^2}{\sum_{i=1}^{N_j} (y_i^{(j)} - \bar{y}_i^{(j)})^2} \quad (3)$$

R^2 quantifies the proportion of variance in the observed data that is explained by the model. A value of 1 indicates perfect explanation, while a value of 0 indicates performance no better than predicting the mean. A negative value implies the model performs worse than the mean baseline. Its interpretation can be misleading with autocorrelated or non-stationary data, necessitating its use alongside other metrics and residual analysis.

The Mean Squared Error (MSE) for station j is defined as:

$$MSE^{(j)} = \frac{1}{N_j} \sum_{i=1}^{N_j} (y_i^{(j)} - \hat{y}_i^{(j)})^2 \quad (4)$$

MSE measures the average of the squared errors. By squaring the differences, it places a higher penalty on large prediction errors (outliers), making it sensitive to extreme values. The result is in squared units of the target variable, which can be converted to the original scale by taking the square root (Root Mean Squared Error).

2.5. Prediction Model

The final model selection was based on a quantitative evaluation of the three previous metrics so that each of the four stations used the model that met the best performance criteria at the local level. The selected model was then used to generate final predictions by applying preprocessing procedures identical to those used in the training stage (handling missing values, transformation/scaling, encoding, etc.) to avoid information leakage. All prediction outputs were stored per station along with reproducibility metadata (model version, hyperparameters, seed, timestamp, and data preprocessing scheme).

3. Results and Discussion

This section presents a series of meteorological data processing results and evaluations of the prediction models used in the study. The results are presented starting from the description of the dataset and preprocessing process, followed by a comparison of the performance of the three machine learning models tested, namely Lasso Regression, XGBoost Regression, and Gradient Boosting Regression. A comprehensive analysis of all MSS stations is also explained to determine the best model for predicting daily rainfall based on historical data.

3.1. Data Descriptions and Final Output

Daily weather data from four MSS observation stations was successfully collected and combined into a single integrated dataset for further analysis. All data is available in numerical form without the need for digitization or extraction from graphs, allowing for direct processing. Although MSS provides historical data since 1980, each station has a different observation period; some stations have complete records from the beginning, while others only started operating in certain years, such as 2009, 2011, or later. This difference in the length of the time series causes variations in the number of observations between stations. After all the data was combined, a series of pre-processing steps were carried out, including cleaning invalid values, imputing missing values using monthly medians, standardizing date formats, converting data types, detecting and handling outliers, and adding time features. The result of these steps was a clean, structured historical dataset ready for use in daily rainfall prediction modeling.

3.2. Comparison of Prediction Models

Three machine learning models were compared in this study, namely Lasso Regression, XGBoost Regression, and Gradient Boosting Regression. The three models were trained and evaluated separately at each of the 4 MSS weather stations using the MSE, MAPE, and R^2 metrics. To illustrate their performance, the evaluation results at several representative stations are shown in Table 1.

Table 1. Evaluation Matriks of MSS Stasiun

Lasso Regression			
Stasiun	MAPE	MSE	R^2
Paya Lebar	5.64811E+12	0.0043	0.999
Ulu Pandan	5.54134E+12	0.0041	0.999
Marine Parade	7.85774E+12	0.0054	0.999
Jurong Island	6.74489E+12	0.004967275	0.999
XGBoost			
Stasiun	MAPE	MSE	R^2
Paya Lebar	6.48444E+11	0.2181	0.998
Ulu Pandan	8.78358E+11	0.0844	0.937
Marine Parade	4.03721E+11	0.3758	0.957
Jurong Island	6.45747E+11	0.2643	0.970
Gradient Boosting Regression			
Stasiun	MAPE	MSE	R^2
Paya Lebar	5.77692E+12	0.0868	0.999
Ulu Pandan	6.72351E+12	0.0588	0.999
Marine Parade	5.93121E+12	0.0751	0.999
Jurong Island	5.40598E+12	0.1456	0.995

Based on Table 1 and the complete evaluation results for all stations, Lasso Regression showed the most consistent and superior performance compared to XGBoost and Gradient Boosting in all observation areas. The Lasso model produced lower MSE values and higher R^2 values at almost all stations, demonstrating a better ability to explain daily

rainfall variability. The superiority of Lasso is also evident from its stability at stations with long and short data ranges, proving this model to be the most adaptive to the varying characteristics of meteorological data in Singapore.

3.3. Determination of the Best Model and Analysis of Results

Overall analysis of the four stations shows that Lasso Regression is the best and most stable model, outperforming the other two models at all locations. The nature of Lasso, which combines L_1 regularization, provides

the ability to suppress irrelevant parameters, resulting in a simpler but still accurate model. This is very important in meteorological datasets that have multicollinearity between variables, such as the close relationship between daily temperature, sub-hourly rainfall, and wind speed.

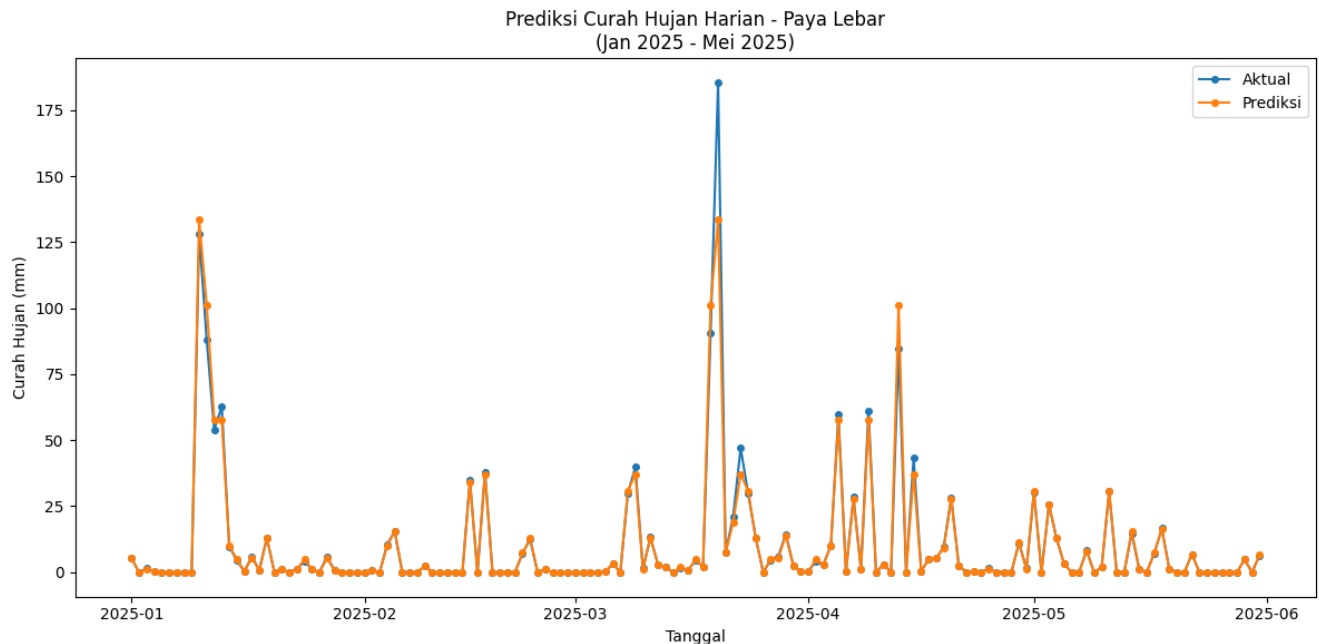


Figure 1. Comparison of Model Prediction Results vs. Actual Data (Paya Lebar Station)

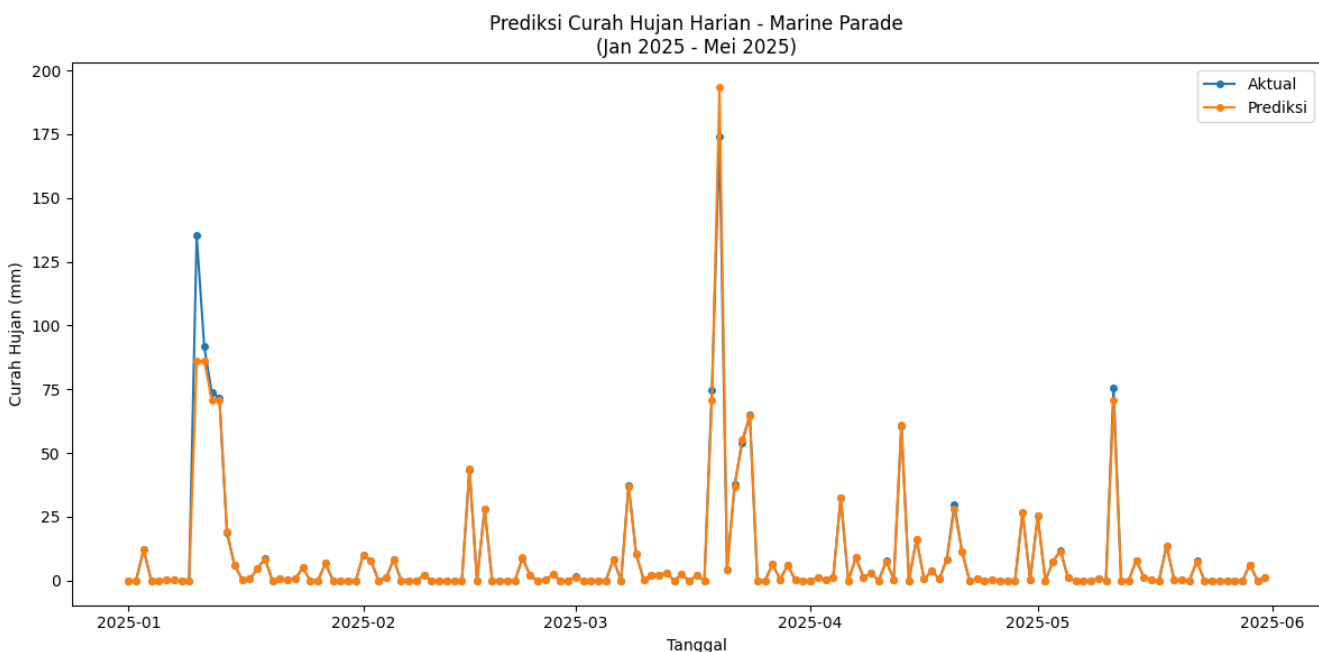


Figure 2. Comparison of Model Prediction Results vs. Actual Data (Marine Parade Station)

Visualization of the comparison between actual and predicted values at stations such as Paya Lebar and Marine Parade shows that Lasso Regression is able to follow the temporal pattern of rainfall quite well, including rainfall peaks and day-to-day fluctuations. These visual results support quantitative evaluation and confirm that Lasso is the most

suitable model for predicting daily rainfall in Singapore based on historical meteorological data.

3.4. Result Interpretations

The results show that Lasso Regression provides the best performance in predicting daily rainfall at all MSS observation stations. This advantage can be explained by

the L_1 regularization mechanism possessed by Lasso, where the model automatically reduces the coefficients of less relevant variables to zero. This approach is particularly effective for meteorological datasets that tend to have high multicollinearity, such as strong relationships between sub-hourly rainfall, daily temperature, and wind speed. By reducing model complexity, Lasso is able to maintain prediction stability and avoid overfitting, which is often a problem with non-linear models such as XGBoost or Gradient Boosting when applied to data with less pronounced variability structures. Visual results at several stations confirm Lasso's ability to follow temporal patterns and rainfall peaks, supporting the conclusion that this model is most suitable for modeling daily rainfall dynamics in Singapore.

3.5. Benefits and the Application of Research

This research provides several important benefits, both methodologically and practically. From a methodological perspective, this study shows that daily rainfall modeling can be performed effectively using historical numerical data from MSS without the need for additional digitization, thereby streamlining the model development process. Testing at four stations shows that regularization-based approaches such as Lasso Regression can produce stable models that are easy to apply in regions with varying meteorological characteristics. In terms of application, the results of this study have the potential to support improvements in Singapore's operational weather prediction system, particularly for flood mitigation, urban water management, and early warning of extreme weather. In addition, this approach can be replicated by meteorological agencies in other countries, including BMKG in Indonesia, because the data format and modeling process are general and can be applied in various geographical contexts.

3.6. Limitations and Recommendations

This study has several limitations that need to be considered. First, although the MSS data is relatively complete and structured, variations in the length of time series between stations can affect model homogeneity and limit long-term generalization. Second, imputation of missing values using monthly medians was chosen to maintain seasonal patterns, but this method may not be optimal for capturing extreme events or weather anomalies. Third, this study only compares three types of models, so it does not cover deep learning-based approaches or advanced time-series models such as LSTM, GRU, or probabilistic forecasting. For further research, model development can be directed towards evaluating more complex architectures, integrating spatial data such as radar or satellite, and exploring model interpretability techniques to understand the influence of each meteorological variable in greater depth.

4. Conclusions

This study aims to develop and evaluate a daily rainfall prediction model using historical meteorological data from four MSS observation stations in Singapore. Through a series of pre-processing and testing of three machine learning models, namely Lasso Regression, XGBoost Regression, and Gradient Boosting Regression, the results

showed that Lasso Regression was the most accurate and stable model across all stations. The L_1 regularization applied in Lasso provided the ability to reduce multicollinearity between variables while producing a simpler and more generalizable model. These findings answer the research question that daily rainfall modeling can be done effectively using a regularized regression approach on numerical meteorological data.

The implications of these results are quite broad for the development of rainfall prediction systems in tropical regions. Stable and easy-to-implement models such as Lasso Regression can be used as basic components in early warning systems and urban flood risk management, especially in Singapore, which faces high rainfall variability on a local scale. Furthermore, this approach can be replicated in other countries, including Indonesia, given the similarity in data structure and climate characteristics. Overall, this study emphasizes the importance of utilizing structured historical meteorological data to build accurate, efficient prediction models that can support efforts to mitigate the impacts of extreme weather in urban areas.

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Informed Consent Statement: Not applicable. Since this study does not involve human subjects, informed consent is not required.

Data Availability Statement: The meteorological datasets analyzed in this study are publicly accessible through the Meteorological Service Singapore (MSS) at: <https://www.weather.gov.sg/climate-historical-daily>. Additional processed data files and modeling scripts can be provided upon reasonable request to the corresponding author.

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